$$\boldsymbol{r}_i^{(-k)} = \boldsymbol{Y}_i - \widehat{\boldsymbol{m}}^{(-k)}(Z_i), \qquad V_i^{\text{avg}} = \frac{\left|\mathbf{1}^\top \boldsymbol{r}_i/12\right|}{\sqrt{c^\top \widehat{\Sigma}_R c}}, \quad c = \mathbf{1}/12,$$

$$\widehat{p}_j^{\,w} = \frac{\omega(Z_j) + \sum_i \omega(Z_i)\mathbb{I}\{V_i \geq V_j\}}{\omega(Z_j) + \sum_i \omega(Z_i)}, \quad \text{retain } j \iff \widehat{p}_j^{\,w} > \gamma = 0.10.$$

**Two-sided:** $V$ uses $|\cdot|$.